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RESEARCH-ARTICLE

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CHAO DENG, Guangdong University of Science and Technology, Dongguan, Guangdong, China

JIAO PENG, Guangdong University of Science and Technology, Dongguan, Guangdong, China

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Application of LSTM Model based on Particle Swarm Optimization Algorithm in Stock Market Trend Prediction

Chao Deng

Guangdong University of Science and Technology
Dongguan, Guangdong, China
dengchao@gdust.edu.cn

Jiao Peng*

Guangdong University of Science and Technology
Dongguan, Guangdong, China
pengjiao@gdust.edu.cn

Abstract

The stock index reflects the development of national economy and is a dynamic, complex, nonlinear system. Accurately predicting future trends in the stock market has important practical significance for regulators, investment institutions, and individuals. With the rapid development of machine learning and deep learning, using time series models, neural networks, machine learning algorithms and other models to predict future trends in the stock market has become a hot research topic. Although Long Short-Term Memory (LSTM) network model has good memory characteristics and is suitable for predicting stock time series data, it lacks the ability to explain final decisions obtained by the model as well as being affected by human factors. Therefore, this study combines particle swarm optimization algorithm with LSTM model to take optimal position of particles as optimal parameters for LSTM neural network prediction of stock data in order to improve accuracy.

CCS Concepts

• **Design and analysis of algorithms** → Mathematical optimization; Non-parametric optimization; Developmental representations.

Keywords

Particle swarm algorithm (PSO), Long short-term memory network model (LSTM), Optimization, forecast

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1 Introduction

The stock market plays an increasingly significant role in the global economy as an indicator and early warning system of national economic development and change. The fluctuation of stock data is closely related to various factors such as macroeconomic situation,

legislative and regulatory framework, political events, corporate financial conditions, policies, investor psychology, public opinion guidance, etc., resulting in high volatility and irregularity of stock prices. Therefore, accurate prediction of stock price changes has become a primary concern for many scholars in this field. In reality, with the integration of computer technology and the financial industry, there is a growing body of research on stock index prediction. Stock forecasting provides guidance and reference value for investment direction for trading institutions; it also promotes market maturity to some extent by providing investors with a certain direction when purchasing stocks to protect their interests. As a result, it has broad application prospects.

The structure of the stock market is complex and easily influenced by external factors while the stock data exhibits nonlinearity and instability to some extent. In earlier research stages limited by computer technology capabilities and imperfect data systems, predicting stock indices mainly relied on traditional econometric models based on historical quotation data. However, with rapid advancements in deep learning models over time, neural network models have gradually been applied to predict future prices and trends in the stock market. Based on numerous scholarly findings [1], LSTM neural network model has demonstrated good performance in predicting stock prices. Like other neural network models though the effectiveness of LSTM model heavily relies on its network structure selection which poses challenges towards improving forecasting ability and accuracy.

1.1 Related work

In the process of designing the stock forecasting model, the stock market structure is complicated and easy to be affected by external factors, so the stock data has some non-linear and unstable. Therefore, time series model is the most commonly used method in stock forecasting. However, due to the high level of noise and nonlinearity inherent in financial time series data, it is challenging to accurately predict stock price trends using linear forecasting models. With the rapid development of computer science and technology and the increasing maturity of deep learning research, neural network has become a more effective and accurate method for stock prediction based on machine learning, especially in China [2]. Machine learning algorithms are able to automatically extract rules from data and use those rules to predict unknown data. In the era of big data, neural networks, support vector machines, decision trees and other methods are widely used in the field of stock forecasting [3].

The deep learning model has more powerful learning ability and adaptive ability, so the deep learning model can better predict and analyze the nonlinear system, and is widely used in the field of

*Corresponding author.



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stock prediction. Long short-term memory network model LSTM adds input gate, output gate and forgetting gate on the basis of recurrent neural network, which can effectively deal with problems such as data forgetting and gradient explosion in recurrent neural network [4].

1.2 Motivation and contribution

The network parameters of the LSTM model usually depend on the selection of human experience, and the number of nodes in different hidden layers and the learning rate will affect the final fitting effect of the network [5]. Therefore, in this study, PSO algorithm was used to optimize the number of hidden layer neurons and dropout rate in LSTM network, so as to reduce the negative impact of parameter selection on the network. The parameters of LSTM model were optimized by PSO, and the mean square error MSE and the square sum of weights MSW were used as the objective functions of PSO algorithm to evaluate the fitness of each particle. By updating particle velocity and position, the optimal number of hidden layer neurons and dropout rate can be found for the network prediction of the next layer.

The main innovations and contributions of this work include:

(1) Innovative application of hybrid model: By combining particle swarm optimization (PSO) algorithm with Long Short-Term Memory (LSTM) neural network, this study proposes a new hybrid model for stock index prediction. This model combines the advantages of two powerful computational tools, namely PSO's ability for global optimization and LSTM's ability for processing time series data.

(2) Improvement of parameter optimization strategy: Traditionally, the setting of neural network parameters such as the number of neurons often relies on empirical Settings or tedious manual adjustments. In this study, the PSO algorithm is used to automatically find the optimal parameter combination, which reduces the dependence of the model on manual experience and may find a more accurate model configuration.

(3) Enhance prediction accuracy: The mean square error (MSE) and the sum of squares of weights are taken as optimization objectives, and the network parameters are iteratively updated to promote the model to gradually improve its prediction performance in the course of multiple training, which is expected to improve the accuracy of stock index prediction.

2 Literature review

Time series model is the most commonly used method in stock forecasting, and it is also one of the commonly used tools to predict stock market changes. Zhang and Sun established the ARIMA model to forecast and analyze the Shanghai Composite Index of stocks in the future and found that the model had the best prediction effect in the next four trading days [6]. Liu and Zhang utilized an ARIMA model to test and predict the historical closing price data of selected financial market stocks over a period of two days. They concluded that the maximum discrepancy between the predicted values and actual values for the subsequent three days was no more than 0.04 [7]. KORS and KARAN found that GARCH model has strong predictive performance [8]. MIAH, RAHMAN et al. used Akaike and Bayes criteria to prove that GARCH model is superior

to other models in predicting return series volatility [9]. Fang et al. used the ARMI-GARCH model to study media stock prices and found that the model was more accurate in predicting individual stocks with "peak and fat tail" distribution characteristics than the traditional ARIMA model and GARCH model [10]. The study of Yang and Cao further demonstrated that the prediction error of the ARMI-GARCH would increase with increase of the number of prediction steps, and the accuracy would also get lower and lower [11]. Xu and Liang established the ARMI-GARCH model to carry out the pseudo-merger of the closing price series of Yutong Bus through empirical error analysis, and proved that the model can be applied to the prediction of stock closing price [12]. Xiong and Che brought volatility into the ARMI-Garch model, built the ARMI-GarCH-M model, and proved through experiments that the model would be more accurate in fitting and predicting short-term stock market prices [13].

Neural network modeling is based on the internal relationship of data itself, and has strong self-organization, adaptability, robust learning ability and anti-interference ability. The convergence speed of neural network to extract economic activity knowledge from historical data for stock market analysis exceeds the limitation of traditional quantitative forecasting methods, and provides a new method for stock market modeling and forecasting. Huang et al. used SVM, BP neural network and wavelet neural network models to train and predict stocks [14]. However, their experiments show that direct use of pre-dictions based solely on the BP algorithm leads to unstable returns, slow convergence, and prone to local minima. Therefore, improvements are needed in terms of accuracy and speed. Zou used particle swarm optimization (PSO) algorithm to optimize the BP neural network model [15]. Experimental verification shows that PSO-BP model is more efficient and feasible for stock market prediction than other methods. Zhang and Zhang compared BP neural network and ARMA-GARCH model and found that BP neural network showed higher accuracy in long-term prediction [16]. Yang and Chen proposed that the combination of wavelet neural network (WNN) and ARIMA, an auto-regressive integral moving average model, is more accurate in short-term stock prediction [17]. Ding (2021) compared the prediction of stock price trend by SRIMA model and LSTM model, and concluded that the prediction accuracy based on LSTM model was better through empirical research [18].

Since stock data is characterized by large amount of data, fuzzy information, long memory, nonlinear and non-stationary, and deep learning model has stronger learning ability and adaptive ability, it can better predict and analyze nonlinear system by using deep learning model. LSTM adds input gate, output gate, forget gate on the basis of re-current neural network, which can effectively deal with data forgetting and gradient explosion problems. It has good selectivity, memorability and timing internal influence characteristics, and is very suitable for the prediction of stock price time series [19]. FISCHER and KRAUSS also believed that whether predicting stock prices or stock returns, model prediction effect based on LSTM neural network is better [20]. Analysis of Xiong and Zhao shows that LSTM model is more suitable for this field of financial market prediction, and beats other prediction models by obvious advantages [21].

Although LSTM is a powerful forecasting tool, its network still have some short-comings. First, LSTM lacks an explanatory model

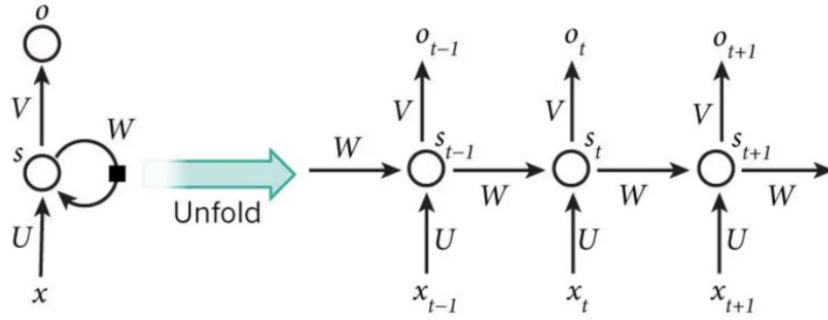


Figure 1: RNN Structure Diagram

to provide specific explanations for the parameters used to achieve the predicted results. Secondly, the parameters of the model are usually defined by experience, which is subjective. Finally, the parameter values of the LSTM network are difficult to determine [22]. Therefore, simply using LSTM model to forecast will greatly reduce the accuracy of model prediction due to the complex and redundant data of influencing factors of stocks.

In this study, particle swarm optimization (PSO) and LSTM network model are combined to complete the data prediction of stock index, so as to improve the prediction accuracy.

3 Terminology introduction

3.1 Neural network

A Neural network is a kind of arithmetic model, which is composed of neuron nodes connected with each other. Each neuron is a computing unit, and each connection between two neurons represents the weight associated with it.

Neural network is divided into Feed forward Neural Network (FNN) and feedback neural network. FNN is the most basic type of neural network. The signal propagates one-way from the input to the output layer, there is no loop, and the information can only flow forward. This model is suitable for supervised learning of numerical data, but it cannot be used together with sequential data and cannot handle image data well.

The output of feedback neural network is not only related to the current input and the weight of the network, but also related to the input before the network, which has strong associative memory ability and optimization computing ability. Common model structures include: Recurrent Neural Network (RNN), LSTM, etc. RNN is the most commonly used feedback neural network, which can process data with sequence changes. In this way, a bidirectional cycle can be formed, and time information can be extracted in the cycle, which has the effect of short-term memory. RNN neurons can receive information from the previous neuron and the outside, process it, and then transmit it to the next neuron, which makes it widely used in time series tasks. Figure 1 shows the network structure of RNN.

By combining time series data with neural network, RNN can retain the influence of previous data on the present. However, with the increase of time series, it is found that RNN is weak in processing long time series. Moreover, with the increase of time series and

depth of model, it is impossible to avoid the problems of gradient disappearing and gradient explosion when calculating gradient, which need to be improved [23]. LSTM can solve these problems. It designed neurons of each layer into a structure with multiple “gates”, so that the error can be directly and intact transmitted to the next layer in the process of propagation without completely disappearing, so it has good convergence [24].

3.2 Long Short-Term Memory Network

The structure of the Long Short-Term Memory network (LSTM) also has an input layer, a hidden layer and an output layer, as shown in Figure 2. There are many storage units in the hidden layer. Each storage unit consists of input gate, forget gate and output gate, which can keep and adjust the state of the storage unit.

The forget gate is calculated by sigmoid, which outputs a value between 0 and 1, so as to decide whether to retain or partially retain the information learned last time, as shown in formula (1).

$$f_t = \sigma(W_f \times [h_{t-1}, x_t] + b_f) \quad (1)$$

Where, f_t the value range of is 0 to 1, the smaller the value, the more forgotten, otherwise, the less.

The input gate is composed of two parts, sigmoid and tanh, to control the retained information. The input gate simply determines what useful information needs to be added to the network's storage unit C_t . The first step is to calculate the activation value i_t for the input gate, then calculate the candidate value $\tilde{C}_t$ that might be added to the cell state, and finally calculate the updated C_t . (2), (3) and (4) show the specific calculation formula.

$$i_t = \sigma(W_i \times [h_{t-1}, x_t] + b_i) \quad (2)$$

$$\tilde{C}_t = \tanh(W_c \times [h_{t-1}, x_t] + b_c) \quad (3)$$

$$C_t = f_t \times C_{t-1} + i_t \times \tilde{C}_t \quad (4)$$

The output gate scales the C_t value to between -1 and 1, and then calculates it with the sigmoid function to get the final result, as shown in formulas (5) and (6).

$$o_t = \sigma(W_o \times [h_{t-1}, x_t] + b_o) \quad (5)$$

$$h_t = o_t \times \tanh(C_t) \quad (6)$$

Although the division of tasks of the three gates is different, the nature used in the operation process is the same. The data information is selected through sigmoid network layer, and the

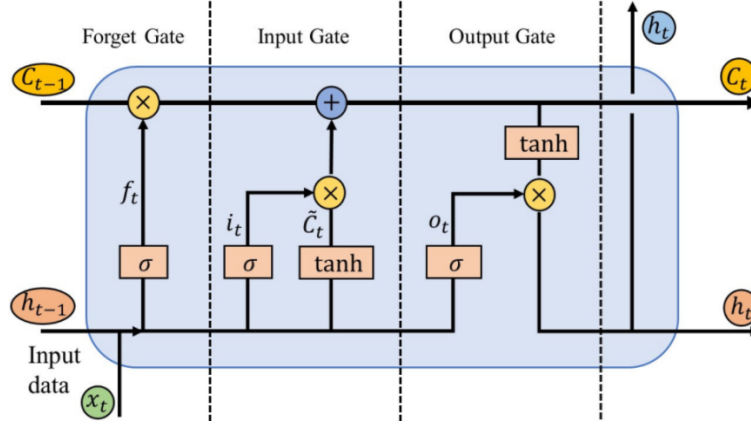


Figure 2: LSTM Structure Diagram

data information is updated through tanh network layer, so as to achieve the final effect cooperatively.

In LSTM neural networks, the selection of key parameters is very important in the process of model construction. Parameters such as the number of neurons, the proportion of neuron disconnection, and the number of samples entering the LSTM network each time will affect the result of neural network prediction. Therefore, the data of hidden layer neurons, the proportion of neuron disconnection and the number of samples entering LSTM network each time are analyzed as parameters to be optimized in this study, so as to accurately predict and help investors better understand the changes of the stock market, so as to make reasonable investment.

3.3 Particle Swarm Optimization Algorithm

Particle Swarm Optimization (PSO), which originated from the study of bird predation behavior, is a global search algorithm for optimal values. The optimization process of PSO algorithm can be compared to the foraging process of birds. Each individual in the flock is equivalent to each particle in the swarm, and the food sought by the birds is the optimal solution sought by the particles [25].

Suppose that in a D-dimensional search space, there are N particles in a particle swarm, each of which represents a potential solution. The position of particle in D dimensional space is \$(X_{i1}, X_{i2}, \dots, X_{id})\$, the speed is \$(V_{i1}, V_{i2}, \dots, V_{id})\$, the optimal position is \$(P_{i1}, P_{i2}, \dots, P_{id})\$, that is, the global optimal value is \$P_{gbest} = (P_{g1}, P_{g2}, \dots, P_{gd})\$. The core of PSO algorithm is to make particles gradually approach the optimal position by constantly adjusting the speed and position of particles. The update of particle swarm in position and speed can be expressed as formula (7) and (8).

$$V_{id}^{t+1} = wV_{id}^t + c_1r_1(P_{id}^t - X_{id}^t) + c_2r_2(P_{gd}^t - X_{id}^t) \quad (7)$$

$$X_{id}^{t+1} = X_{id}^t + V_{id}^{t+1}T \quad (8)$$

$$P_{ibest}^{t+1} = \begin{cases} X_i^{t+1}, f(X_i^{t+1}) \leq f(P_{ibest}^t) \\ P_{ibest}^t, f(X_i^{t+1}) > f(P_{ibest}^t) \end{cases} \quad (9)$$

$$f(P_{gbest}^t) = \min [f(P_{ibest}^t)], i = 1, 2, \dots, N \quad (10)$$

The global optimization ability of PSO algorithm is strong, but the local optimization ability is insufficient and the generalization ability is poor. Therefore, Liu Jun et al. proposed to combine the mean square error of training samples and the sum of squares of weights as the fitness function [26]. This method has good stability, high prediction accuracy and significantly enhanced generalization ability. Therefore, this study uses the combination of sample mean square error MSE and weight squared and MSW as the objective function of PSO algorithm.

$$MSE = \frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2 \quad (11)$$

$$MSW = \frac{1}{N} \sum_{i=1}^N w_i^2 \quad (12)$$

$$f(x) = \gamma \times MSE + (1 - \gamma) \times MSW \quad (13)$$

4 ESTABLISH PSO-LSTM MODEL

The LSTM model is suitable for processing time series data, but it is difficult to determine parameters such as the number of neurons in the network. In general, it is based on experience and has a great deal of randomness. PSO algorithm is used to optimize the number of neurons and the learning rate of LSTM neural network to improve the prediction accuracy of LSTM model. By using the PSO algorithm, the position information of the optimal individual is used as the initial neural number, output rate and batch size of LSTM to train the LSTM neural network, so as to improve the prediction accuracy of the LSTM model [27].

Figure 3 shows the structure of the PSO-LSTM model, which uses the PSO algorithm to search the parameters of LSTM.

(1) Initialize PSO parameters to determine particle swarm size, iteration number, search dimension, initial inertia weight and learning factor.

(2) Set the position and speed of the particle. Generate a random particle \$Xi=(h1,h2,d,b)\$, \$h1\$ and \$h2\$ represent the number of neurons at layer 1 and Layer 2 of the neural network, \$d\$ is the dropout ratio,

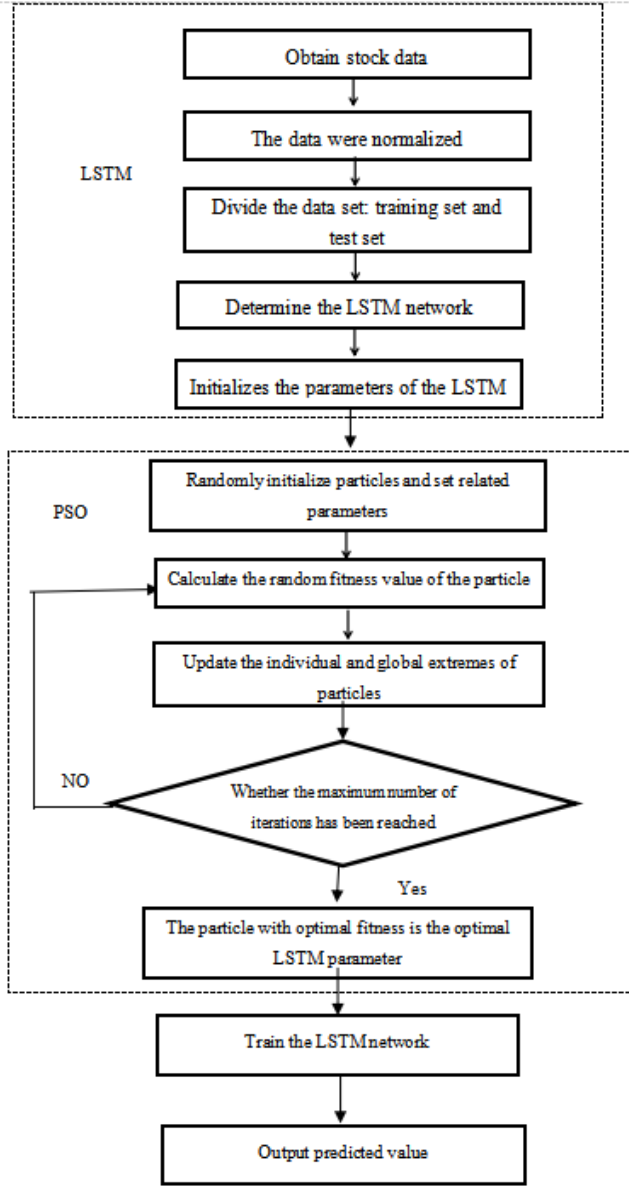


Figure 3: Structure of PSO-LSTM Model

and b is the size of `batch_size`. Particle speed $V_i = (V1, V2, V3, V4)$, the velocity of each dimension is $[0, 1]$.

(3) Determine other parameters of LSTM, including the number of hidden layers and output layers, the number of output layer neurons, loss function, optimizer, etc.

(4) The fitness function of particle swarm optimization is determined.

For each particle, the LSTM model is built using its current hyperparameter configuration. The LSTM model is trained on the training set and the prediction error is calculated on the validation set. The square of MSE and weight and MSW are combined as the fitness value of the particle.

For each particle, the current fitness value is compared with the historical optimal fitness value, and the individual optimal position is updated. Compare the fitness values of all particles and update the global optimal position.

According to the speed update formula of PSO algorithm, the speed and position of particles are adjusted.

(5) Iteration. Calculate the globally optimal particle position. The global optimal particle position was mapped to the number of neurons, dropout rate and batch-size of LSTM neural network, and the optimal parameters were obtained.

(6) Make predictions. The LSTM neural network is trained by using the global optimal particle position to determine the optimal parameters, and the final prediction results are output.

5 Experimental analyses

5.1 Data analysis and processing

Based on the stock data of Shanghai Pudong Development Bank 600000.SH, this study analyzed 1,481 pieces of data, including trading date, opening price, high price, low price, closing price, trading volume, market value, turnover rate, Price-to-Earnings (PE)Ratio, Price-to-Book (PB) Ratio and other indicators. This study intends to forecast the rate of return, so an “earn_rate” column is added to store the daily rate of return. The data header information is shown in Figure 4.

5.2 Time series diagram of stock return rate

With time as the horizontal coordinate and daily return rate as the vertical coordinate, it is presented in the form of a line chart, through which the basic situation of stock return rate changing over time can be observed, as shown in Figure 5.

5.3 Correlation analysis of stock indicators

The function `Pandas.scatter_matrix()` is used to calculate correlations between a number of indicators selected from the data and correlate the representative indicator data correlations together to create a scatter plot, where the diagonal is the histogram of each representative indicator data. Figure 6 shows that there is a linear positive correlation between the opening price, closing price, high price and low price of a stock. Figure 7 shows the correlation between stock volume and turnover. It can be analyzed from Figure 7 that there is a positive linear relationship between volume and turnover rate. When the volume rises, the turnover rate also rises, and vice versa.

5.4 Data analysis and processing

1. Normalized processing. Normalization is to remove the dimensional influence between data indicators, so as to solve the comparability between data indicators. After the initial data has been standardized, each data index is at the same quantity level, and a comprehensive comparison can be made on the data. The function used for normalization is the `MinMaxScaler` function, and the range of values is defined as $[-1, 1]$.

```

sc = MinMaxScaler(feature_range = (-1, 1))
data_scaled = sc.fit_transform(data)
  
```

	t-date	open	high	low	close	volume	m-value \
date							
2013-01-04	20130104	10.10	10.24	9.93	10.02	261436810	1.870000e+11
2013-01-07	20130107	10.05	10.33	10.02	10.32	248200915	1.930000e+11
2013-01-08	20130108	10.34	10.34	10.08	10.12	154851189	1.890000e+11
2013-01-09	20130109	10.09	10.18	9.99	10.14	175906181	1.890000e+11
2013-01-10	20130110	10.11	10.20	10.00	10.04	121226794	1.870000e+11

	turnover	pe	pb	earn_rate
date				
2013-01-04	1.40	5.467378	1.000320	0.029940
2013-01-07	1.33	5.631072	1.030270	0.029940
2013-01-08	0.83	5.521943	1.010303	-0.019380
2013-01-09	0.94	5.532856	1.012300	0.001976
2013-01-10	0.65	5.478291	1.002317	-0.009862

Figure 4: Header information for data

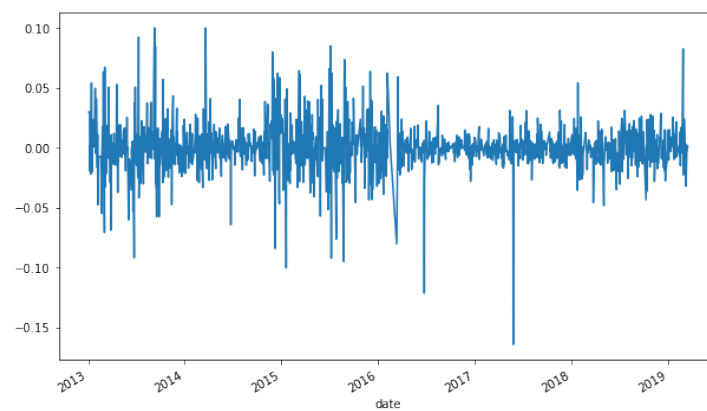


Figure 5: Time series of stock returns

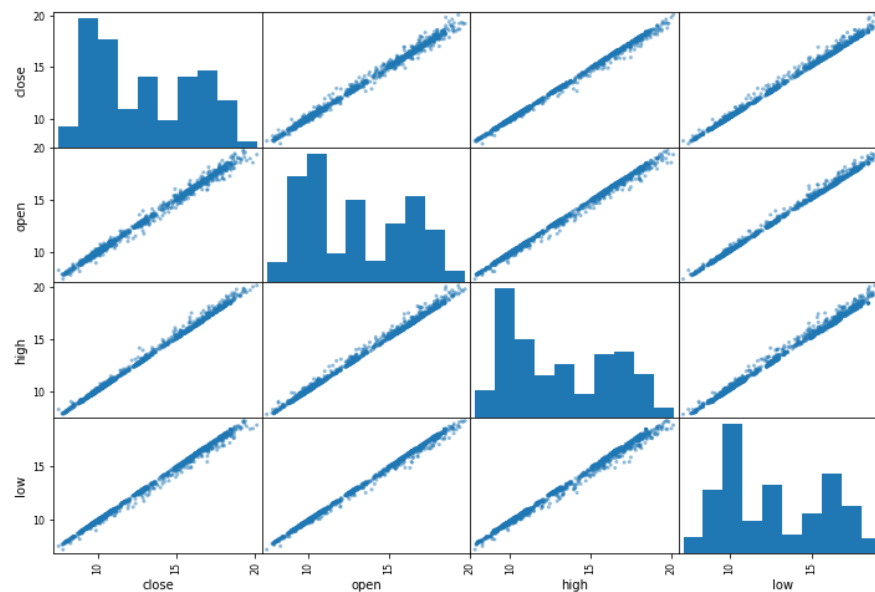


Figure 6: Correlation of stock index

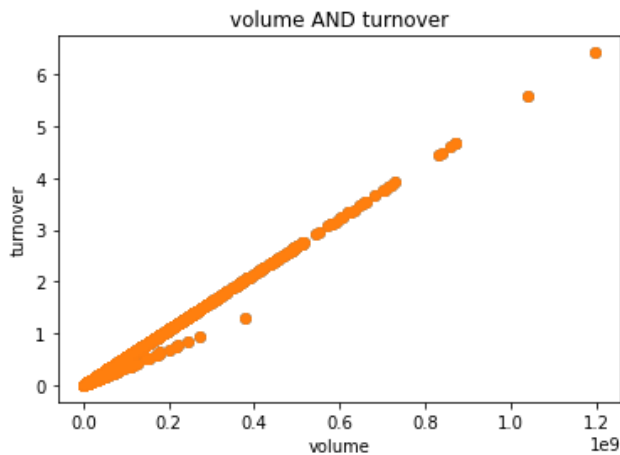


Figure 7: Correlation between stock volume and turnover rate

2. Divide the data set. Using the `train_test_split()` function, we used 80% of the data set as training data and 20% as testing data.

3. Construct the PSO-LSTM network model and analyze the results. Initialize particle swarm parameters, and set the parameters of PSO algorithm as follows: particle number N is 2, iteration number t is 20, search dimension D is 4, learning factor $=1.5$, inertia weight $w=0.5$. Initialize the position and speed of the particle. Set the dropout rate to a random number between 0 and 1. Set the batch-size value to $[20,100]$. Particle speed $V_i = (V1, V2, V3, V4)$, the speed of each dimension is $[0,1)$.

Two hidden layers and one output layer of the LSTM neural network are set. The output layer has one neuron, which is responsible for the output of the predicted rate of return. The loss function is MSE and the optimizer is Adam.

The LSTM neural network is constructed with the initialized particle swarm parameters, and the sum of squares of the MSE and weights of the actual and tested values of the training set is used as the objective function of the PSO, where the value of γ is set to 0.9.

Calculate the position and fitness values of the particle in each iteration. Global optimal particle position was mapped into the number of neurons, dropout rate and batch-size of the LSTM, and the LSTM neural network is trained with the optimal parameters and the final prediction results are obtained.

Model training was carried out according to the above steps, the test set was input into the model, and the prediction results were output, as shown in Figure 8. The green solid line is the predicted value, and red line is the true value. The predicted stock return is in line with the trend of the true value. The MAE is 0.046087, the MSE is 0.09029, and the RMSE is 0.095022. The error value is relatively small, and the results show that the LSTM prediction model optimized by PSO algorithm has a better fitting degree, which is consistent with the conclusion proposed by Xie et al. [28]. Because the stock market is greatly affected by external factors, so from the technical point of view, the accuracy of prediction is relatively high.

6 Conclusion

This study realizes a model that can be used in quantitative investment practice teaching to forecast stocks. The model is based on LSTM neural network to forecast stock returns. In the process of model study, to solve the problem that the prediction results of LSTM neural network are easy to be affected by network structural parameters, researchers chose PSO to optimize the parameters of the network. The prediction effect of LSTM model optimized based on PSO algorithm is good, which has certain research value and practicability. The stock market itself is a complex system, which is affected by many factors such as national political situation, policy, finance and enterprise situation. The stock index prediction based on PSO-LSTM algorithm is only a data prediction from a technical perspective. Therefore, when the external environment changes greatly, such as changes in enterprise conditions, there will often be relatively large deviations when making predictions based on past data. Therefore, when making stock investment, investors should not only refer to the results of data model analysis, but also consider the influence of external economic, policy, social and cultural factors.

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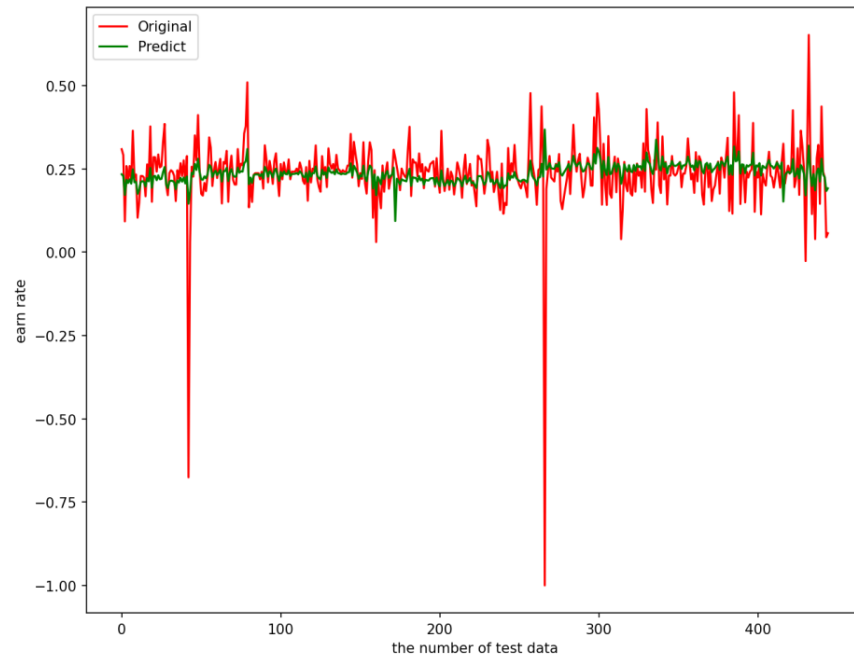


Figure 8: Prediction Results

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